

ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

DIPARTIMENTO DI SCIENZE ECONOMICHE

DEPARTMENT OF ECONOMICS

Corso di Laurea - *First-cycle/Bachelor's degree*

In

ECONOMICS AND FINANCE

At the Edge of Conservation: Livelihoods and Household Welfare in Maasai-Associated Areas of Northern Tanzania

PRESENTATA DA / PRESENTED BY

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Sessione di laurea di Luglio / *Graduation session of* July

Anno Accademico – *Academic year* 2025 / 2026

Abstract

This paper studies whether households located in Maasai-associated areas near Tanzanian game reserves experienced different welfare and livelihood changes after conservation-relevant reform periods. The empirical setting combines repeated cross-sections from the Tanzania National Panel Survey with a spatially constructed measure of Maasai-associated locality exposure. The main sample is restricted to households within 75 kilometres of a game-reserve boundary. The preferred treatment identifies households located within 25 kilometres of Maasai-associated localities; a narrower 10-kilometre treatment is used as a high-intensity robustness check. The results do not show a large and systematic decline in average consumption. They do show a clearer increase in poverty after 2009 and stronger evidence of pastoral mechanisms: cattle decline, cattle exit, higher treatment costs, higher livestock expenses, animal disease and later increases in milk-production value. The interpretation is not simple herd collapse. It is a costly transformation of pastoral livelihoods under conservation-related pressure. The paper also contributes methodologically. Because ethnicity and local language are not directly observed in modern Tanzanian household surveys, Maasai association is reconstructed spatially from locality evidence rather than from unavailable individual ethnic identifiers.

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1 Introduction

Protected areas are central to biodiversity policy, but they are also politically and economically contested spaces. In low-income rural settings, the land around protected areas is rarely empty. It is used for grazing, seasonal mobility, cultivation, water access, trade and social reproduction. Conservation policy therefore raises a development question as well as an environmental question: who bears the local costs of conservation and through which livelihood margins do those costs appear?

This paper studies this question in Tanzania, focusing on households in Maasai-associated areas near game reserves. Maasai communities are historically associated with pastoralism, cattle-based livelihoods and seasonal mobility across semi-arid areas of northern Tanzania and southern Kenya. These features make the Maasai case especially relevant for studying conservation. Restrictions on land access, grazing mobility and settlement can affect pastoral livelihoods even when they do not immediately appear as large drops in measured consumption.

The research question is:

> Among households located in a local reserve-boundary risk set, did post-2009 conservation exposure differentially affect households in Maasai-associated areas?

The answer is nuanced. The analysis does not find strong evidence of broad consumption collapse. The consumption coefficients are negative in some specifications, but they are not statistically precise. The poverty indicator increases after 2009. More importantly, the mechanism results are stronger than the aggregate consumption results. Cattle dynamics deteriorate after 2009: the probability of cattle decline rises, cattle exit rises, large-ruminant treatment costs and livestock expenses rise sharply, later outcomes also point to disease incidence and higher milk-production value. These patterns are more consistent with costly livestock adjustment than with immediate livelihood destruction.

The analysis is deliberately conservative. It does not claim that conservation exposure is randomly assigned. It uses repeated-cross-section regressions with wave fixed effects, district fixed effects, household controls, a linear control for distance to the nearest game-reserve boundary and standard errors clustered by enumeration area. The core comparison is local: households must be within 75 kilometres of a game-reserve boundary. This reduces the risk of comparing Maasai-associated households with unrelated rural economies far from the conservation frontier.

2 Historical and policy motivation

2.1 Conservation and local development

The economics of conservation is shaped by distributional trade-offs. Protected areas may preserve biodiversity and landscapes, but they may also restrict rural land use. This is especially important where livelihoods depend on extensive land access rather than fixed, privately titled plots. Pastoralism is one of those systems.

The development-economics literature has increasingly treated conservation as both an environmental and a welfare question. Studies such as Andam et al. (2010) and Ferraro and Hanauer (2011) evaluate whether protected areas reduce poverty or generate local trade-offs. Other work stresses that local effects are heterogeneous and can depend on enforcement, land tenure, market access and baseline livelihood systems. A protected area may have little average effect on consumption while still changing the way households produce income. This is the empirical intuition of the present paper.

Tanzania is a particularly relevant case. It has a large protected-area estate and a long history of wildlife conservation. At the same time, the areas around reserves are inhabited or used by rural communities. Maasai-associated areas are central to this debate because pastoral livelihoods rely on cattle mobility, grazing access and water access. If conservation policy constrains mobility or changes land access, the first effects may appear in herd stress, disease exposure, veterinary costs or milk production, not necessarily in immediate consumption.

2.2 Policy timing: 2009 and 2013

The empirical timing is built around two policy-relevant periods. The first is the post-2009 period. In 2009, Tanzania adopted the Wildlife Conservation Act No. 5 of 2009, which replaced the earlier wildlife legislation and strengthened the legal framework for wildlife protection, protected-area governance, hunting regulation and enforcement authority (United Republic of Tanzania, 2009). For the purposes of this paper, the 2009 indicator should not be interpreted as the effect of one legal clause. It captures the first survey period after a broader legal consolidation of wildlife governance.

The second timing margin is the additional post-2013 period. The 2013 marker is interpreted as an enforcement-pressure period rather than as a single statute. In 2013, anti-poaching enforcement became especially salient in Tanzania. Contemporary human-rights reporting describes a controversial anti-poaching campaign in which members of parliament and civil-society actors alleged excessive force, cattle seizure and killing and

burning of pastoralists' houses in relation to grazing on protected land; President Kikwete dismissed four ministers in December 2013 after parliamentary concern over the operation (U.S. Department of State, 2014). This context makes 2013 a plausible marker for an intensification of enforcement pressure around protected areas. The analysis therefore separates an initial post-2009 shift from an additional post-2013 shift. This does not imply perfectly dated causality. It provides a transparent way to describe whether outcomes changed after the legal reform period and whether they shifted again after the enforcement-intensification period.

2.3 Climate narratives and the reason for an NDVI check

Tanzanian policy documents correctly emphasize climate vulnerability. The National Adaptation Programme of Action, prepared by the Vice President's Office, describes Tanzania's economy as dependent on natural resources and rain-fed agriculture and identifies severe and recurrent droughts as a major development risk (United Republic of Tanzania, 2007). Agriculture, including livestock, is explicitly central to rural livelihoods in the same document. This official climate narrative is important because worsening pastoral conditions can easily be attributed to drought, vegetation decline or dry-weather shocks.

This paper does not deny the importance of climate stress. Instead, it asks whether the estimated Maasai-associated differential changes are simply absorbed by local vegetation conditions. For this reason, NDVI is used as a robustness control. NDVI is an additional satellite-based environmental covariate, available only for EA-wave observations with valid spatial information and potentially measured with noise due to jittered survey coordinates and satellite-composite limitations. It is therefore preferable not to include it in the baseline specification, but to test whether the main coefficients are robust to controlling for local vegetation conditions.

The NDVI construction is harmonized across survey rounds. NDVI is first sampled at all available georeferenced survey points and then aggregated to the enumeration-area-by-wave level before being merged into the regression dataset. This harmonizes the spatial level of the satellite control across survey rounds, since some NPS waves provide household-level coordinates while others provide cluster-level coordinates. Sampling is conducted at a 1-kilometre scale (`scale=1000`) because the public NPS coordinates are spatially displaced. A much finer grid would create false precision; a coarser measure would risk smoothing away relevant environmental variation. The NDVI robustness specification keeps the same wave and district fixed effects as the baseline and adds the EA-wave NDVI control.

The aim is not to prove that drought does not matter. The aim is narrower: to test whether

the main welfare and livestock-stress results disappear once local vegetation conditions are controlled for. They do not. In fact, the poverty coefficient becomes larger in the NDVI specification, which suggests that the Maasai-associated welfare effect is not simply a greenness or dry-weather artifact.

2.4 The Nyerere legacy and the measurement problem

A central difficulty is that ethnicity is not directly observed in the survey data. This is not a minor technical inconvenience. It is a historically meaningful feature of Tanzanian data.

After independence, Julius Nyerere's government pursued a project of national integration. Tanzania is ethnolinguistically diverse, but post-independence state-building placed strong emphasis on national identity and the use of Swahili as a unifying language. A major interpretation in the political-economy literature is that Tanzanian nation-building reduced the political salience of ethnic divisions (Miguel, 2004). The empirical consequence is clear: contemporary household surveys are not designed around direct ethnic identification.

The treatment in this paper is therefore area-based. A household is classified as exposed when it is located near localities identified as Maasai-associated from documentary and spatial evidence. This is not the same as individual ethnic self-identification. Some households in Maasai-associated places may not be Maasai, and some Maasai households may live outside classified localities. But the estimand is intentionally local: it asks whether places associated with Maasai pastoral livelihoods changed differently after conservation-relevant reform periods.

This measurement strategy is imperfect, but it is also the methodological contribution of the paper. It allows an ethnicity-relevant question to be studied in a country where direct ethnicity and language variables are unavailable in modern household surveys.

I did not find a close development-economics paper that does exactly the same operation, namely building a local ethnic-exposure layer through documentary scraping, geocoding and manual validation. The closest precedents are methodological rather than identical. First, several economics papers use historical ethnic homelands or ethnographic maps to assign group-level cultural exposure when individual ethnicity is not directly observed (Nunn and Wantchekon, 2011; Michalopoulos, 2012; Alesina et al., 2013). Second, political methodology and public-policy research often imputes race or ethnicity from imperfect auxiliary information such as surnames and residential geography (Imai and Khanna, 2016; Rosenman et al., 2022). The common logic is that identity-relevant exposure can be studied using external spatial or textual information when direct self-identification is unavailable. The present paper applies that logic to a Tanzanian setting and keeps the interpretation explicitly area-based rather than individual-ethnic.

3 Data and spatial construction

3.1 Input datasets

The analysis combines five types of data. First, household outcomes come from the Tanzania National Panel Survey, accessed through the World Bank NPS data files and the associated NBS documentation (World Bank, 2024; National Bureau of Statistics, 2016, 2022). Second, the harmonisation workflow builds on the EPAR LSMS data-dissemination codebase for Tanzania NPS files (EPAR, n.d.). Third, the administrative spatial backbone uses the 2022 Tanzania ward boundaries from the National Bureau of Statistics (National Bureau of Statistics, 2022b). Fourth, game-reserve boundaries are taken from the Africa GeoPortal game-reserve shapefile and are used to compute distance to the nearest reserve boundary (Africa GeoPortal, n.d.). Fifth, the broad reference polygon for ethnic settlement is the GeoEPR/EPR family of geocoded ethnic-settlement data (Vogt et al., 2015; GeoEPR, 2021). The Maasai-associated treatment layer is then constructed by combining documentary evidence on localities with GIS validation.

3.2 Household survey data

The empirical analysis uses the Tanzania National Panel Survey (NPS). The NPS is a repeated household survey containing consumption, poverty, livestock ownership, livestock sales, agricultural activity, income sources, household demographics and geographic information. The main analysis uses Waves 1–4 (respectively: 2008-2009, 2010-2011, 2012-2013, 2014-2015). Wave 5 (2020-2022) is not included in the main specification because current Wave 5 household coordinates are not publicly available. A Wave 5 extension would require assigning later outcomes to Wave 4 baseline spatial exposure, which changes the estimand.

The unit of observation is a household-wave. Outcomes are grouped into welfare, herd dynamics, livestock costs and productivity, agricultural mechanisms and income reallocation. The final interpretation focuses on a small set of outcomes: consumption, poverty, cattle dynamics, large-ruminant treatment costs, livestock expenses, disease, milk value produced and livestock income share.

3.3 Game-reserve risk set

Each household is assigned its distance to the nearest game-reserve boundary. The main sample is restricted to households within 75 kilometres of a game-reserve boundary. This creates a local conservation risk set. The purpose is to avoid comparing households close

to reserves with households in distant rural economies that are not plausibly exposed to the same conservation frontier.

The game-reserve shapefile is used to define the reserve-boundary distance. Figure 1 shows the spatial distribution of the game-reserve polygons used in the construction of the risk set.

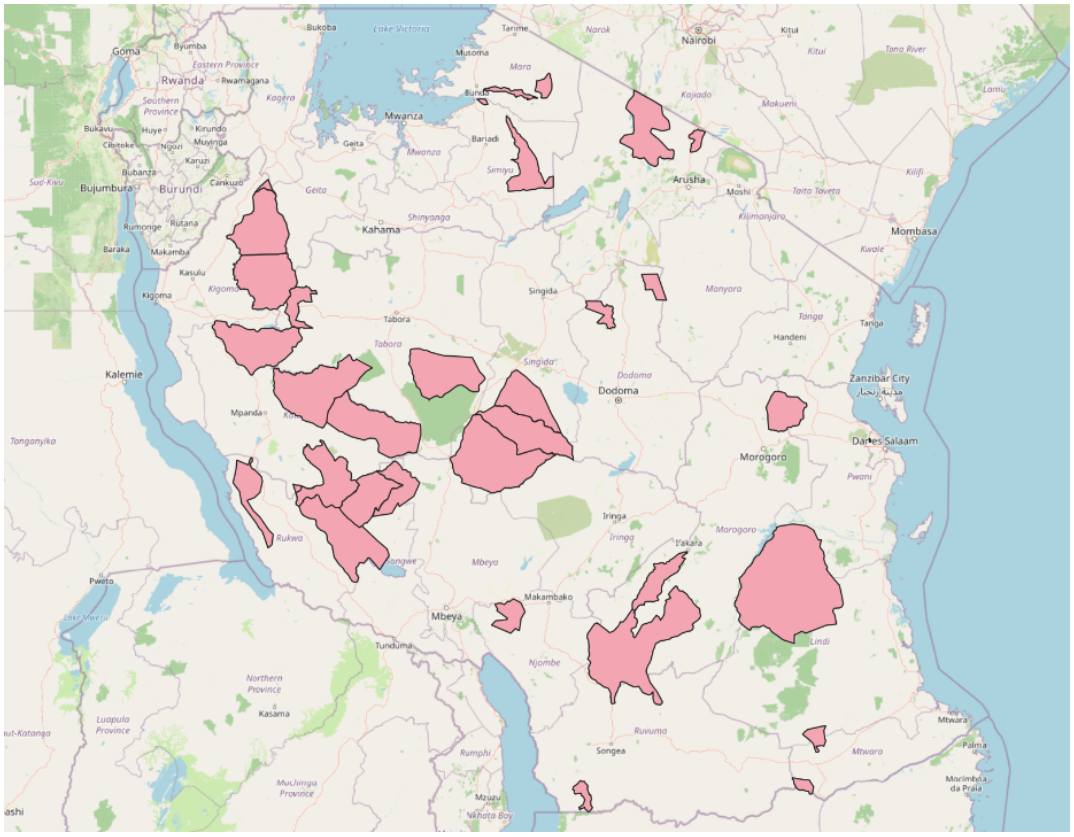


Figure 1: Game-reserve polygons used to construct the local reserve-boundary risk set. Notes: the figure is a QGIS screenshot of the game-reserve shapefile used to compute distance to the nearest reserve boundary.

3.4 Maasai-associated exposure

The preferred treatment variable is an indicator for being within 25 kilometres of a Maasai-associated locality. The choice of 25 kilometres balances two concerns. A very narrow treatment captures a high-intensity exposure group, but support becomes thin. A broader treatment improves support while remaining spatially local. The 10-kilometre treatment is therefore kept as a high-intensity robustness check, not as the preferred baseline. It is relevant to recall that household coordinates are offset for privacy reasons, thus justifying the use of a cushion around Maasai-associated localities.

Table 1 reports the key support comparison. Inside the 75-kilometre risk set, the 25-

kilometre treatment identifies 630 treated household-wave observations across 74 treated enumeration-area clusters. The 10-kilometre treatment identifies only 184 treated observations across 28 treated clusters. This difference is the main reason for using the 25-kilometre treatment as the baseline and the 10-kilometre treatment as a robustness check.

Table 1: Treatment support inside the 75 km reserve-boundary risk set

Treatment definition	Treated observations	Treated EA clusters	Interpretation
25 km Maasai-associated exposure	630	74	Main treatment
10 km Maasai-associated exposure	184	28	High-intensity robustness check

Notes: Counts refer to the W1–W4 local risk set within 75 km of a game-reserve boundary. The broader 25 km treatment is preferred because it offers substantially better support while still identifying Maasai-associated local exposure.

Figure 2 shows the spatial overlap between the constructed Maasai-associated exposure layer and the broad GeoEPR ethnic-settlement polygon. The treatment layer was built from documentary evidence and manually validated by the author. AI assistance was used to speed up documentary search and extraction from academic research, official documents, association materials and news sources, but the GIS layer and final classification were manually checked. The figure is a QGIS visualization of the constructed spatial data.

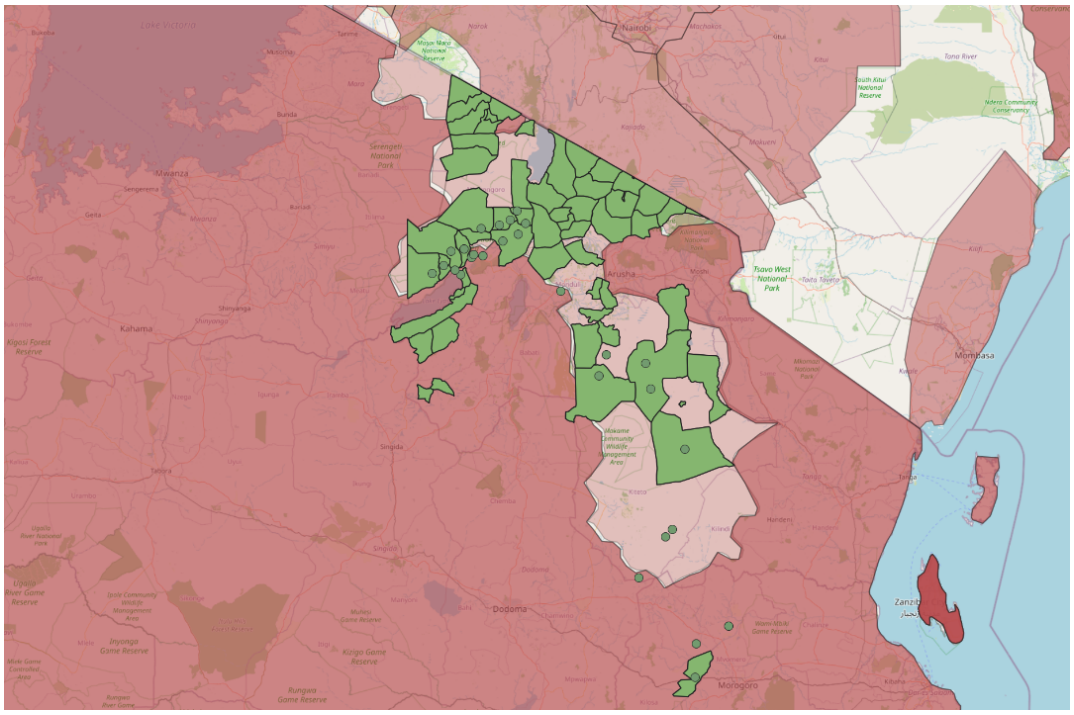


Figure 2: Constructed Maasai-associated exposure layer and GeoEPR reference polygon. Notes: green polygons indicate the manually validated Maasai-associated spatial layer used to construct treatment; the light pink underlying area shows the broad GeoEPR reference polygon. Points indicate locations in the constructed dataset whose polygon is unavailable; they are identified by the name of an institution (e.g., a school or a hospital). The figure is a QGIS screenshot.

4 Empirical strategy

4.1 Two-shock specification

The main specification is a repeated-cross-section difference-in-differences style regression. It allows for two post periods: a first post-2009 shift and an additional post-2013 shift. The estimated equation is:

$$\begin{aligned} Y_{iwt} = \alpha + \beta_1(T_i \times Post2009_w) + \beta_2(T_i \times Post2013_w) + \delta T_i \\ + X'_{iwt}\gamma + \lambda_w + \mu_d + \theta Distance_i + \rho Inside_i + \varepsilon_{iwt}. \end{aligned} \tag{1}$$

where Y_{iwt} is an outcome for household i in wave w, T_i is the Maasai-associated area indicator, $Post2009_w$ is equal to one for survey observations after the 2009 Wildlife Conservation Act period, and $Post2013_w$ is equal to one for observations in the later enforcement-intensification period. The model includes household controls, wave fixed effects, district fixed effects, a linear distance control to the nearest reserve boundary and an indicator for being inside a game reserve. Standard errors are clustered at enumeration-area level.

The coefficient β_1 measures the first differential shift after the 2009 legal-consolidation period. The coefficient β_2 measures the additional shift after the 2013 enforcement-pressure period. The total post-2013 effect is the linear combination $\beta_1 + \beta_2$. The two indicators are therefore period markers rather than claims that all effects occur exactly in those calendar years.

4.2 Event-study interpretation and identification

The event-study specification estimates treated-by-wave interactions relative to Wave 3. It is useful for timing, but it is not a perfect parallel-trends test. There is only one true pre-2009 wave. Therefore, the event study can show whether the main two-shock pattern is driven by specific waves, but it cannot prove the identifying assumption.

The estimated event-study equation is:

$$\begin{aligned} Y_{iwt} = \alpha + \sum_{k \in \{1,2,4\}} \pi_k \left(T_i \times \mathbf{1}\{Wave_w = k\}\right) + \delta T_i \\ + X'_{iwt}\gamma + \lambda_w + \mu_d + \theta Distance_i + \rho Inside_i + \varepsilon_{iwt}. \end{aligned} \tag{2}$$

Wave 3 is omitted and is therefore the reference wave. Each coefficient π_k measures the treated-control difference in wave k relative to the treated-control difference in Wave 3, conditional on the same controls, fixed effects, inside-reserve status and reserve-boundary distance used in the main specification.

The design is not randomized. Maasai-associated areas differ from other areas in geography, pastoral history and local economic structure. The empirical strategy reduces this concern through a local reserve-boundary risk set, district fixed effects, wave fixed effects, household controls, inside-reserve status and reserve-boundary distance. The estimates should be interpreted as conditional differential changes, not as definitive causal effects.

5 Main results

5.1 Welfare and herd dynamics

Table 2 reports the main welfare and herd-dynamics results. The consumption coefficients are negative in the post-2009 period, but they are not statistically precise in the main specification. The poverty result is clearer: the poverty indicator increases by about 9.5 percentage points after 2009. The wild-cluster-bootstrap p-value is 0.080, so the result is less sharp than the ordinary clustered p-value but remains suggestive.

The herd dynamics are stronger. Net cattle change falls after 2009. The probability of any cattle decline rises by about 17 percentage points, and cattle exit also rises. Cattle loss share increases. These outcomes are directly connected to pastoral livelihoods and therefore provide more informative evidence than aggregate consumption alone. Later, total livestock per adult equivalent increases in the additional post-2013 period. This should not be interpreted mechanically as a welfare improvement: a larger livestock count can coexist with higher disease, higher treatment costs and costly adjustment.

Table 2: Main effects on welfare and herd dynamics

Outcome	Post-2009	Additional post-2013	Total post-2013	Obs.	EA clusters
Log consumption per capita	-0.083 (0.081) [0.300]	-0.025 (0.087) [0.785]	-0.108 (0.120)	8,850	257
Log consumption per adult equivalent	-0.095 (0.079) [0.245]	0.012 (0.085) [0.891]	-0.083 (0.115)	8,850	257
Poverty indicator	0.095** (0.045) [0.080]	-0.031 (0.045) [0.513]	0.065 (0.064)	8,959	261
Net cattle change	-2.045* (1.076) [0.046]	1.497 (1.067) [0.202]	-0.549 (0.444)	5,029	154
Any cattle decline	0.174*** (0.059) [0.007]	-0.068 (0.068) [0.331]	0.106*** (0.036)	5,029	154
Cattle exit	0.079** (0.031) [0.057]	-0.076** (0.031) [0.056]	0.003 (0.021)	1,124	77
Cattle loss share	0.078* (0.041) [0.094]	-0.047 (0.041) [0.303]	0.031 (0.031)	1,124	77
Total livestock per adult equivalent	1.292 (1.751) [0.603]	2.115* (1.213) [0.118]	3.406 (2.233)	6,380	160

Notes: Each cell reports the coefficient, the EA-clustered standard error in parentheses, and, where available, the wild-cluster-bootstrap p-value in square brackets. Significance stars are based on ordinary clustered p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The total post-2013 effect is the linear combination of the two primitive interaction coefficients; wild-bootstrap p-values are not reported for this linear combination. All specifications include wave fixed effects, district fixed effects, household controls, inside-reserve status, and linear distance to the nearest game-reserve boundary.

5.2 Livestock costs, productivity and reallocation

Table 3 reports the main mechanism results. The strongest evidence appears in livestock costs and productivity. Large-ruminant treatment costs increase after 2009 and rise even more in the additional post-2013 period. Livestock expenses also increase after 2013. Large-ruminant disease rises after 2013. Milk value produced rises as well.

These patterns point to costly transformation rather than simple herd collapse. If households were abandoning livestock entirely, one would expect falling milk production and possibly lower treatment costs. Instead, the evidence shows higher costs and higher milk value. A plausible interpretation is that Maasai-associated households remain engaged in livestock production, but production becomes more expensive and more exposed to animal-health stress.

Table 3: Main mechanisms: costs, productivity, reallocation, and agriculture

Outcome	Post-2009	Additional post-2013	Total post-2013	Obs.	EA clusters
	0.979***	3.391***			
	(0.329)	(1.156)	4.370***		
Large-ruminant treatment cost	[0.007]	[0.066]	(1.054)	8,959	261
	0.431	2.682**			
	(0.614)	(1.256)	3.113**		
Livestock expenses	[0.503]	[0.108]	(1.509)	8,959	261
	-0.006	0.152**			
	(0.032)	(0.075)	0.147*		
Large-ruminant disease	[0.958]	[0.141]	(0.081)	8,959	261
	0.310	3.775**			
	(1.087)	(1.682)	4.085***		
Milk value produced	[0.793]	[0.115]	(1.220)	8,444	261
	-0.009	-0.179*			
	(0.060)	(0.108)	-0.188		
Livestock income share	[0.879]	[0.241]	(0.123)	2,944	149
	0.031	0.093			
	(0.306)	(0.341)	0.123		
Hired farm labour days	[0.925]	[0.791]	(0.473)	8,361	256

Notes: Each cell reports the coefficient, the EA-clustered standard error in parentheses, and, where available, the wild-cluster-bootstrap p-value in square brackets. Significance stars are based on ordinary clustered p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The total post-2013 effect is the linear combination of the two primitive interaction coefficients; wild-bootstrap p-values are not reported for this linear combination.

Figure 3 visualizes the cost and productivity mechanism. This is the cleanest graphical summary of the main economic mechanism: post-2009 treatment costs rise and post-2013 livestock expenses, disease, and milk production point to a later phase of costly adjustment.

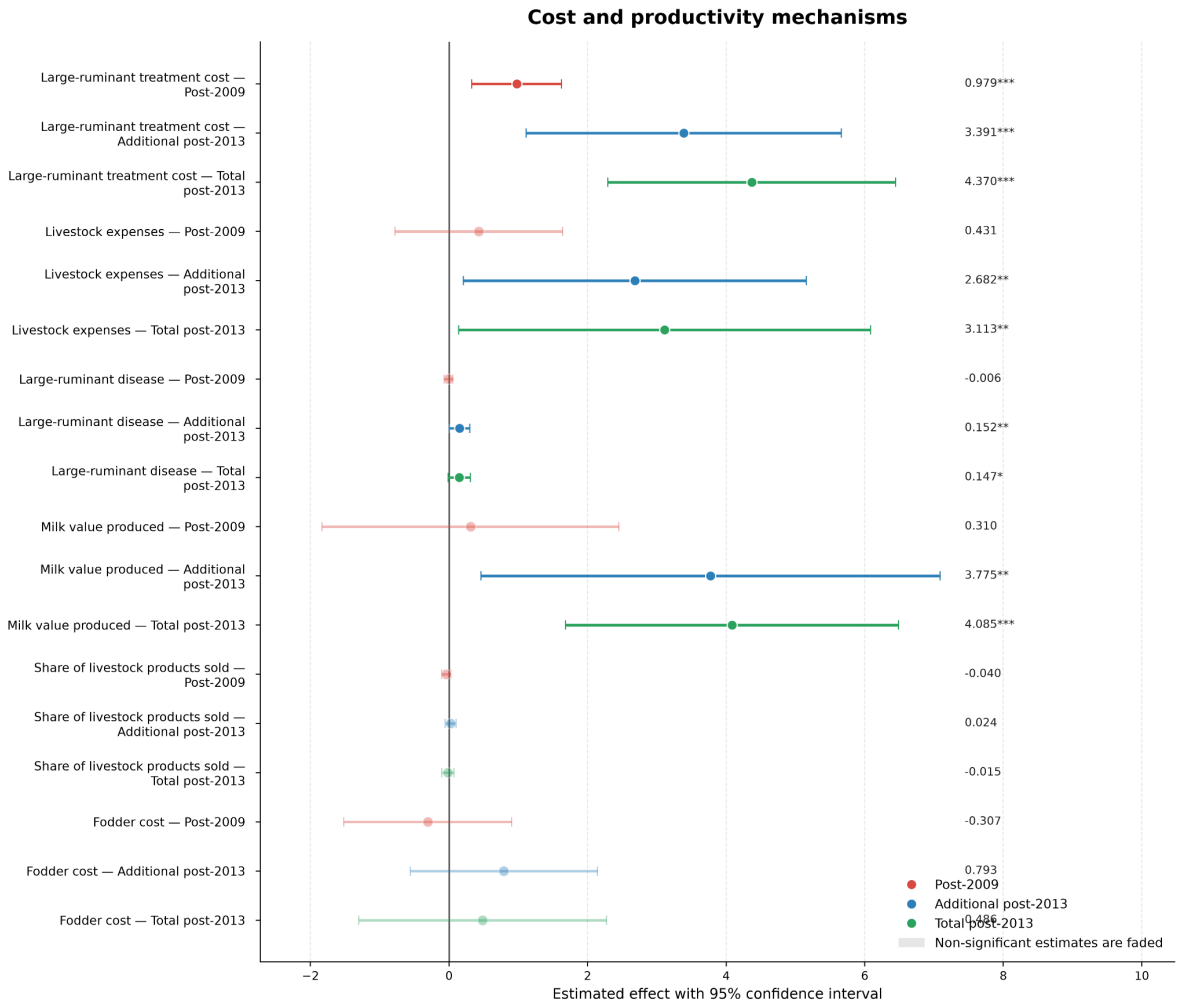


Figure 3: Cost and productivity mechanisms. Notes: points show estimated coefficients from the main specification; horizontal bars show 95 percent confidence intervals. Non-significant estimates are faded. Coefficients are expressed in the units of each outcome and should not be compared mechanically across rows.

5.3 Agricultural mechanisms and income reallocation

The agricultural mechanism is not the central finding. Agricultural participation, land, plot count, family labour, total farm labour and harvest value do not show a systematic pattern in the main specification. This matters because it prevents an overinterpretation of the results as a clean switch from pastoralism to crop agriculture.

Income reallocation is also secondary. The most relevant income result is the decline in livestock income share after 2013. This could reflect lower profitability of livestock production, diversification into other income sources or difficulty converting livestock activity into net income. The evidence does not identify a single destination sector. For this reason, the paper treats income reallocation as a secondary mechanism behind the stronger livestock-cost and herd-dynamics results.

5.4 Event-study dynamics

The event-study figures provide a timing check on the main two-shock estimates. As described in equation 2, Wave 3 is the omitted reference category. The coefficients should therefore be interpreted as treated-control differences in each displayed wave relative to the treated-control difference in Wave 3. The exercise is descriptive rather than a full parallel-trends test, because the survey contains only one true pre-2009 wave.

Figure 4 shows that the welfare outcomes do not follow a clear pattern of immediate consumption collapse. Log consumption per adult equivalent is positive in Wave 1 and close to zero in Waves 2 and 4, with confidence intervals crossing zero throughout. The poverty indicator is more suggestive, moving from a negative coefficient in Wave 1 to a positive coefficient in Wave 2, but the estimates are not precise enough to describe a persistent welfare decline. This is consistent with the main results: aggregate consumption is not the outcome where the evidence is strongest.

The same figure provides clearer evidence on herd dynamics. Net cattle change is essentially flat in Wave 1, becomes sharply negative in Wave 2, and then moves closer to zero in Wave 4. The probability of any cattle decline follows a similar timing pattern: it is negative and imprecise in Wave 1, rises strongly in Wave 2, and becomes smaller and statistically imprecise in Wave 4. The event-study dynamics therefore support the interpretation that the strongest cattle-loss margin is concentrated around the early post-2009 period rather than increasing monotonically across all post-periods.

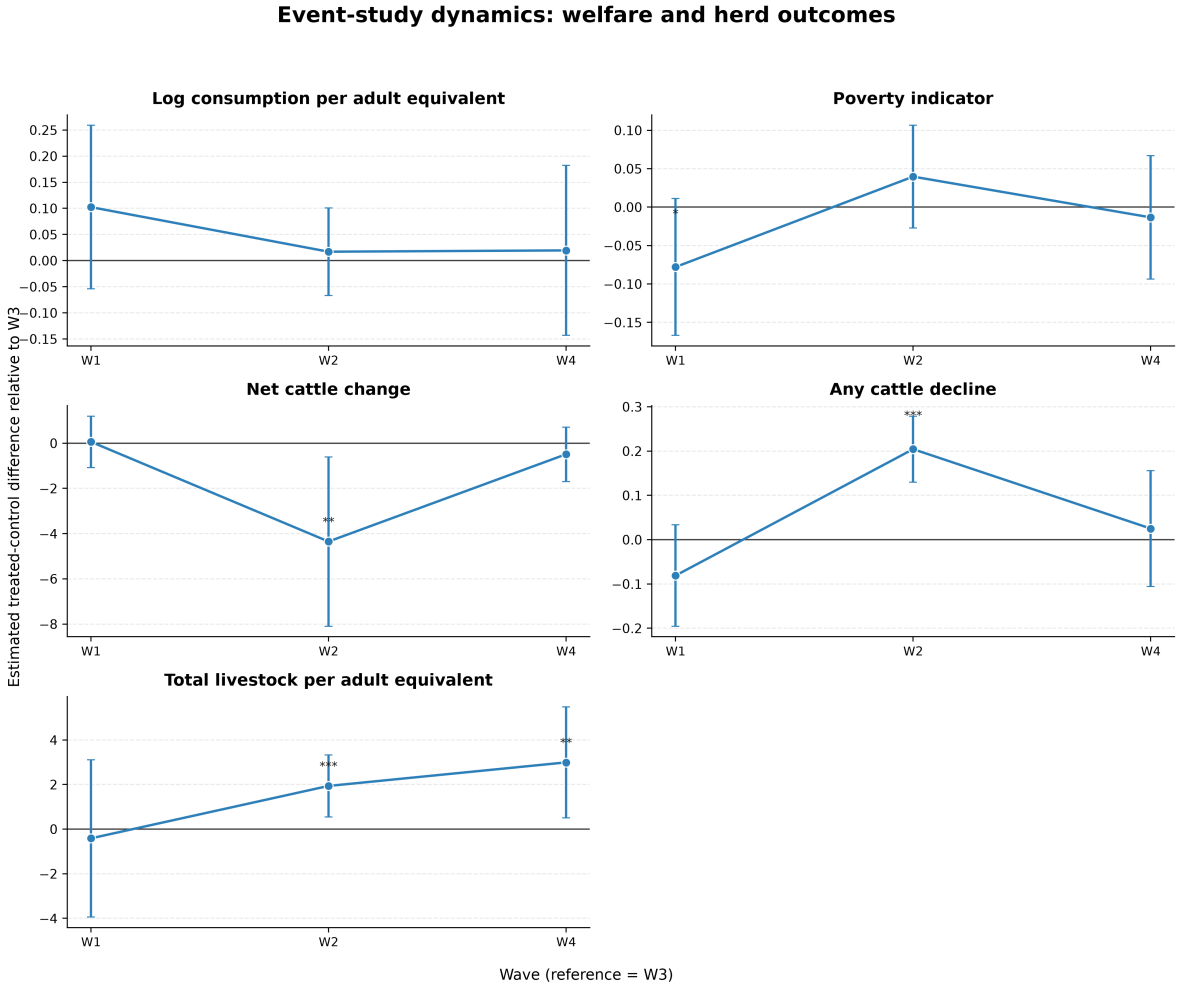


Figure 4: Event-study dynamics: welfare and herd outcomes. Notes: coefficients are treated-by-wave interactions relative to Wave 3. The figure is descriptive and should not be interpreted as a full parallel-trends test, since only Wave 1 is strictly pre-2009. The main pattern is weak for consumption but stronger for herd dynamics, especially net cattle change and cattle decline in Wave 2.

Figure 5 shows the timing of the mechanism outcomes. Large-ruminant treatment costs are negative relative to the Wave 3 reference gap in Waves 1 and 2, but become strongly positive in Wave 4. Livestock expenses and milk value produced also rise in Wave 4. This timing is important because it does not look like a simple abandonment of livestock. Instead, it suggests a later adjustment margin in which households remain engaged in livestock production, but production becomes more expensive and possibly more health-intensive after the enforcement-pressure period. The increase in milk value produced should therefore be interpreted cautiously: it is not necessarily a pure welfare gain, but part of a broader adjustment in which livestock production continues under higher costs.

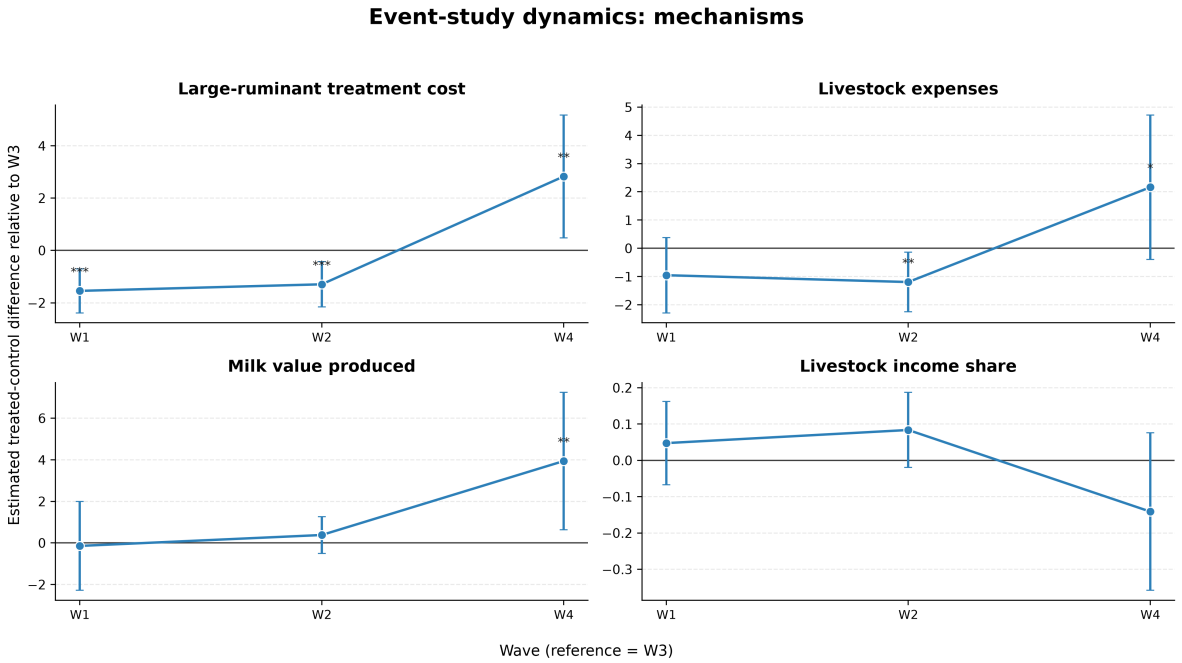


Figure 5: Event-study dynamics: mechanisms. Notes: coefficients are treated-by-wave interactions relative to Wave 3. The figure suggests a later cost and productivity margin: large-ruminant treatment costs, livestock expenses and milk value produced increase most clearly in Wave 4.

6 Robustness checks

Table 4 compares the main specification with three alternatives. The first uses the narrower 10-kilometre treatment. This treatment identifies a smaller and more intensive exposure group, so it is not preferred as the baseline, but it is informative about spatial concentration. The second restricts the reserve-boundary risk set to 50 kilometres while keeping the 25-kilometre treatment. The third adds the NDVI control described above.

The overall pattern is broadly consistent with the main interpretation. The 10-kilometre treatment produces stronger coefficients for several outcomes, which is consistent with spatial concentration around core Maasai-associated localities. The 50-kilometre risk set is more demanding and loses support, but the livestock-stress and cost mechanisms remain visible. The NDVI robustness check is especially important for interpretation. After controlling for local vegetation conditions, the estimated post-2009 poverty coefficient increases from 0.095 to 0.123 and the wild-bootstrap p-value falls to 0.023. Cattle decline and net cattle change also remain stable. This means that controlling for local greenness does not attenuate the welfare-stress result; if anything, it makes the poverty effect clearer. The evidence therefore weakens a purely climate-based explanation in which the Maasai-associated pattern is only a proxy for dry vegetation conditions.

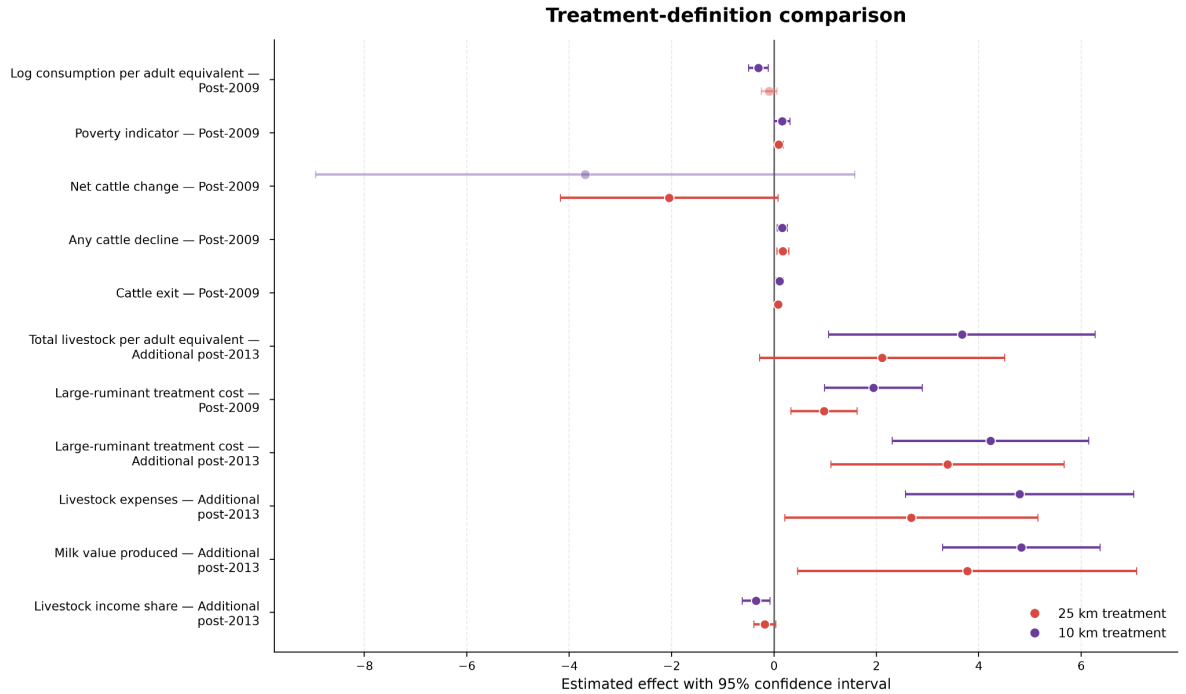


Figure 6: Treatment-definition robustness: 25 km versus 10 km Maasai-associated exposure. Notes: the 25 km treatment is the main treatment because it provides stronger support; the 10 km treatment is interpreted as a high-intensity exposure check.

Table 4: Robustness checks for core coefficients

Outcome and coefficient	Main	10 km treatment	50 km risk set	NDVI control
	0.095**	0.159**	0.040	0.123***
	(0.045)	(0.077)	(0.067)	(0.041)
Poverty indicator – Post-2009	[0.080]	[0.204]	[0.567]	[0.023]
	-0.095	-0.306***	-0.106	-0.153**
	(0.079)	(0.098)	(0.111)	(0.072)
Log consumption per adult equivalent – Post-2009	[0.245]	[0.097]	[0.352]	[0.055]
	-2.045*	-3.686	-4.890**	-2.035*
	(1.076)	(2.666)	(2.402)	(1.052)
Net cattle change – Post-2009	[0.046]	[0.238]	[0.124]	[0.044]
	0.174***	0.163***	0.259***	0.176***
	(0.059)	(0.051)	(0.084)	(0.057)
Any cattle decline – Post-2009	[0.007]	[0.058]	[0.074]	[0.006]
	0.079**	0.111***	0.128***	0.080**
	(0.031)	(0.036)	(0.037)	(0.031)
Cattle exit – Post-2009	[0.057]	[0.058]	[0.133]	[0.061]
	2.115*	3.672***	3.339**	2.230*
	(1.213)	(1.318)	(1.461)	(1.200)
Total livestock per adult equivalent – Additional post-2013	[0.118]	[0.045]	[0.022]	[0.100]
	0.979***	1.941***	0.885**	1.020***
	(0.329)	(0.484)	(0.364)	(0.330)
Large-ruminant treatment cost – Post-2009	[0.007]	[0.052]	[0.029]	[0.005]
	3.391***	4.231***	3.675***	3.352***
	(1.156)	(0.975)	(0.670)	(1.154)
Large-ruminant treatment cost – Additional post-2013	[0.066]	[0.047]	[0.013]	[0.066]
	2.682**	4.798***	4.541***	2.603**
	(1.256)	(1.133)	(0.899)	(1.258)
Livestock expenses – Additional post-2013	[0.108]	[0.058]	[0.006]	[0.115]
	3.775**	4.832***	5.767***	3.707**
	(1.682)	(0.781)	(1.239)	(1.683)
Milk value produced – Additional post-2013	[0.115]	[0.008]	[0.033]	[0.116]
	-0.179*	-0.348**	-0.297*	-0.180*
	(0.108)	(0.137)	(0.153)	(0.108)
Livestock income share – Additional post-2013	[0.241]	[0.136]	[0.050]	[0.242]

Notes: The main specification uses the 25 kilometre Maasai-associated treatment and the 75 kilometre reserve-boundary risk set. The 10 kilometre treatment is a high-intensity exposure check. The 50 kilometre risk set keeps the main 25 kilometre treatment but restricts the local comparison sample. The NDVI column adds an EA-wave NDVI control to the main specification. Cells report coefficients, EA-clustered standard errors in parentheses, and wild-cluster-bootstrap p-values in square brackets where available. Stars are based on ordinary clustered p-values.

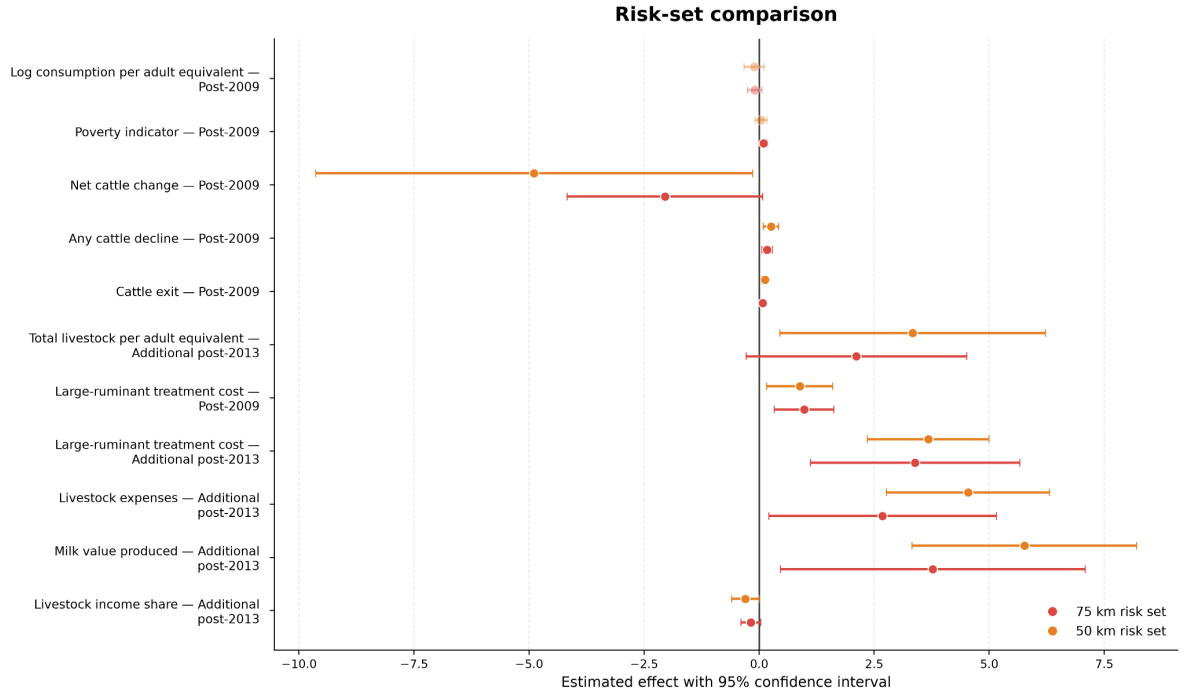


Figure 7: Risk-set robustness: 75 km versus 50 km reserve-boundary sample. Notes: the 50 km risk set is a more demanding local comparison and mechanically reduces the estimating sample.

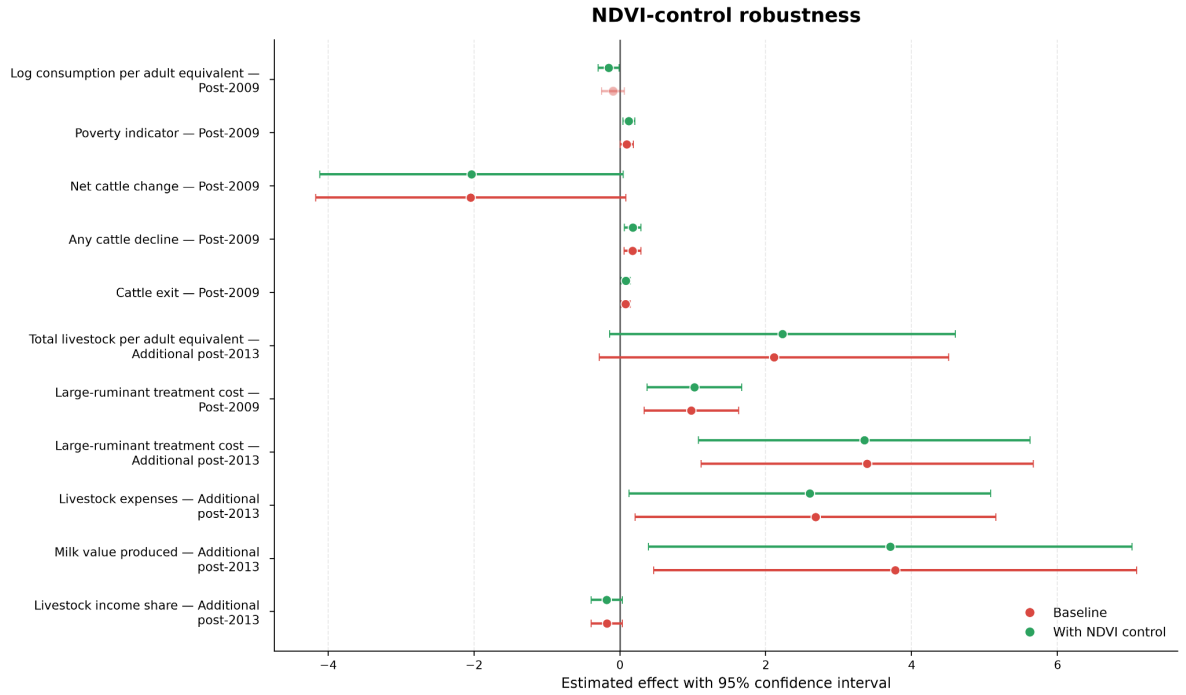


Figure 8: NDVI-control robustness. Notes: the NDVI specification adds an EA-wave vegetation control sampled at 1-kilometre scale and aggregated to enumeration-area-by-wave level. The main welfare and livestock-stress coefficients remain stable. The consumption coefficient also strengthens in the NDVI specification, becoming statistically precise, which further weakens a climate-based confounding interpretation.

7 Limitations

The first limitation is non-random exposure. Maasai-associated areas differ from other areas for reasons that may also affect livelihood trends. The model controls for district fixed effects, wave fixed effects, household characteristics, inside-reserve status and reserve-boundary distance, but unobserved time-varying differences may remain.

The second limitation is treatment measurement. The treatment is locality-based. It does not identify individual ethnicity. This is both a limitation and a contribution. It is a limitation because some households may be misclassified. It is a contribution because it offers a feasible way to study Maasai-associated areas in a survey environment where ethnicity and language are not directly observed.

A related concern is that many locations are classified as Maasai-associated using documentary evidence observed or recorded in a post-treatment period. If locality labels changed because of migration, settlement, administrative reclassification or conflict after the policy period, the treatment could partly reflect post-treatment geography. I reduce this concern by interpreting treatment as area-based Maasai association rather than individual identity, but I cannot eliminate it.

A third limitation is unobserved migration. The repeated-cross-section design observes households in survey waves, not the complete movement histories of people and herds. If conservation pressure caused some households to leave treated areas or induced seasonal relocation that is not captured in the survey, the estimates are conditional on observed household locations and may understate or mischaracterize displacement effects.

A fourth limitation is timing and parallel trends. It is not possible to implement a convincing parallel-trends test for the main post-2009 shift because the NPS provides only one true pre-2009 wave. This is a structural data limitation, not a matter of specification choice. The event-study exercise is therefore descriptive: it shows whether the estimated pattern is concentrated in particular waves, and it is more informative for the additional post-2013 shift than for the initial post-2009 shift. The paper consequently avoids claiming that parallel trends are proven. The post-2009 and post-2013 indicators should be read as reform-period and enforcement-pressure markers, not as perfectly dated policy shocks.

A fifth limitation is satellite measurement. NDVI is useful as an environmental robustness control, but it is not a perfect measure of drought, pasture quality, water access, or livestock mobility. The 1-kilometre scale is appropriate given coordinate jittering, but it also smooths local heterogeneity. The NDVI check therefore tests robustness to local vegetation conditions, not to all climate-related mechanisms.

A final limitation is multiple testing. The analysis includes many outcomes. This is normal

in mechanism-heavy development work, but it increases the risk of over-interpreting isolated significant coefficients. For this reason, the paper focuses on coherent patterns across related outcomes: poverty, cattle decline, cattle exit, treatment costs, disease, milk production and livestock income share.

8 Conclusion

This paper studies how households in Maasai-associated areas near Tanzanian game reserves changed after conservation-relevant reform periods. The results do not support a simple story of immediate consumption collapse. They do support a more specific story of pastoral livelihood stress.

The strongest evidence is not in average consumption. It is in poverty and livestock mechanisms. Maasai-associated households experience higher poverty after 2009, worse cattle dynamics, higher cattle exit, higher large-ruminant treatment costs, higher livestock expenses, higher disease incidence and higher milk-production value. The pattern suggests costly adaptation. Livestock production continues, but it appears to become more expensive and more fragile.

The paper also contributes methodologically. Because ethnicity and language are not directly observed in modern Tanzanian surveys, the analysis reconstructs Maasai association spatially. This is not a perfect substitute for individual identity, but it is appropriate for an area-based research question. It allows the study of a politically important group without relying on unavailable ethnicity variables.

The policy implication is cautious but important. Conservation policy should not be evaluated only through average consumption or aggregate poverty. For pastoral communities, the costs of conservation may appear through animal health, input costs, herd dynamics, and livelihood reallocation. These mechanisms matter because they shape long-run resilience even when short-run consumption effects are muted.

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